



CS 302: Paradigms of Programming

Course Code	CS 302
Course Name	Paradigms of Programming
L-T-P-C	3-0-2-4 3-0-2-4
Prerequisites	IC 150 Computation for Engineers
Intended for	UG
Distribution	Compulsory for CSE; CS elective for EE and ME
Semester	5th
One-time approval	recommended by the 14th BoA (Sep 2016)
Revised	15th BoA (Dec 2016) and 12th Senate (Mar 2017) - credits changed to 3-0-2-4

Course Contents

- **Module I** Lambda Calculus - Syntax, Conversion, Reduction and Normal Order, Church-Rosser Theorem, Order of Evaluation, Currying, Integers, Booleans and Recursion.
- **Module II** Functional Programming - Scheme/Lisp syntax - expressions and functions/procedures, evaluation - naming, environment and the substitution model of function application, Higher-order functions and higher-order programming, Data abstraction.
- **Module III** Object-oriented Programming - Mutable data, modularity and state, Objects, Closures - data structures encapsulated into functions.
- **Module IV** Delayed Evaluation, Laziness and Infinite Data-structures - Stream Programming, Infinite streams, Streams as lazy Lists.
- **Module V** Logic Programming - Deductive Information retrieval, declarative programming - facts and rules, Search trees and Backtracking, Non-deterministic programming, Continuations and back-tracking, Prolog - arithmetic, recursion, cuts and negation, Real-life applications of Prolog. Prolog implementation in Scheme/Lisp - use of continuations and macros.